



HTTP / 1.1 · Conf42 Observability 2026

404

Observability Without Trade-Offs **Not Found**

— until we built **Gigapipe**.

Rita Lopes · founder, Gigapipe

github.com/metrico/gigapipe

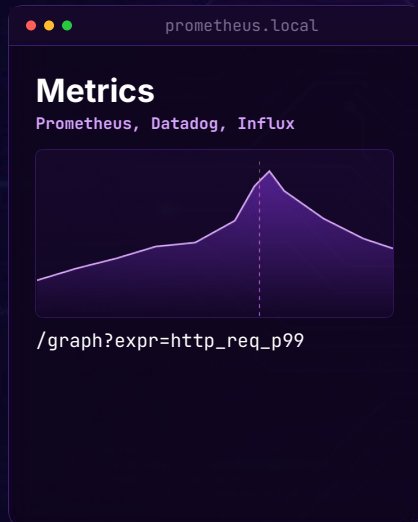
● INCIDENT · P1

It's 3am.

Your phone is ringing.

● LATENCY UP ● ERROR RATE UP ● USERS FEELING IT

The Observability Tax



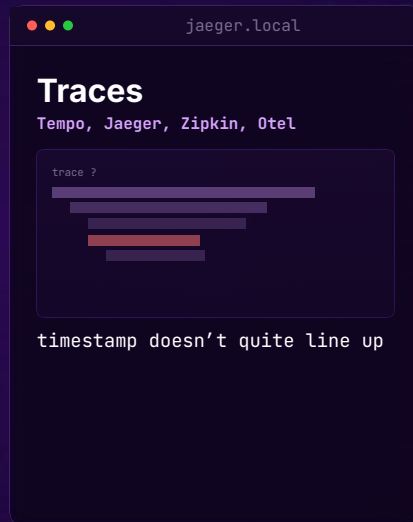
loki.local

Logs

Loki, Otel, Datadog, Elastic, Influx

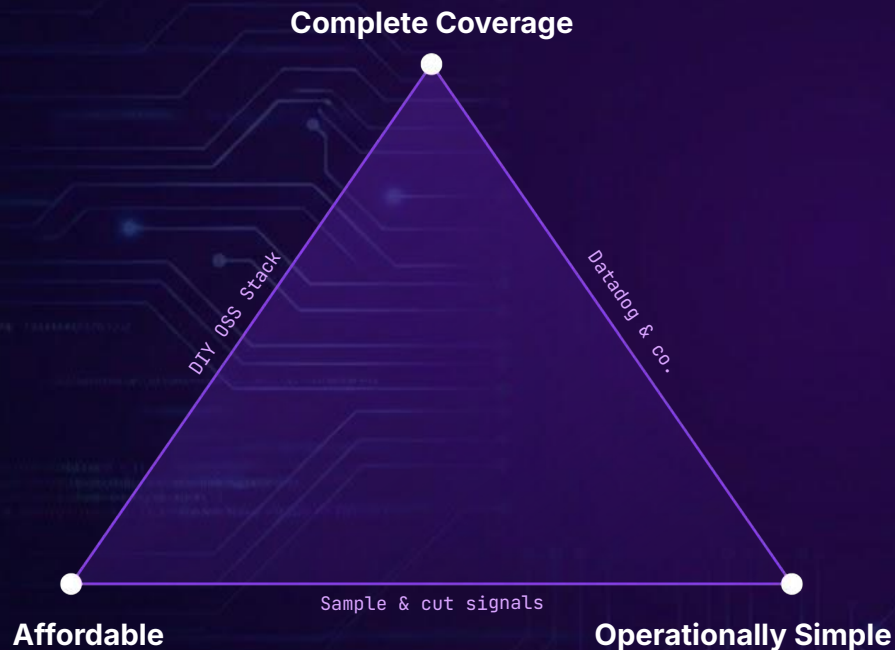
```
02:47:11.221
ERROR svc=payment timeout
02:47:12.503
ERROR svc=payment 5xx
02:47:13.882
ERROR svc=payment ...
```

{job="api"} |= "ERROR"



😓 **jumping between tools and query languages**, correlating everything manually, under pressure, at 3am...
Every signal pays rent: on your time, not just your bill.

The Trade-Off: Pick two. **Never three.**



COMPLETE + AFFORDABLE

You run everything yourself.

A part-time job maintaining four storage backends, four upgrade cycles, four ops runbooks.

COMPLETE + SIMPLE

You go to a vendor.

The bill is... let's say, significant. Pricing-per-event punishes the data you most need to keep.

AFFORDABLE + SIMPLE

You start cutting signals.

Sample traces, drop tier-2 services, shrink retention. Then when there's an incident you will need that data... You're blind.

The cloud bill is the tip.



BELOW THE WATERLINE – INVISIBLE, BUT LARGER

- ◆ Engineer hours maintaining the stack
- ◆ Oncall cognitive load
- ◆ Context switching during incidents
- ◆ Onboarding new SREs to **N** tools
- ◆ Time lost to manual correlation
- ◆ **Delayed MTTR** on every incident

These don't show up in any budget. **They compound.**

AT SOME POINT WE STOPPED ACCEPTING THIS

What if all four signals lived in one place?

STORAGE

**Same backend.
Same time index.**

CORRELATION

**Native, not manual.
One click, not five tabs.**

AGENTS & DASHBOARDS

**Existing tools.
Zero retraining.**

We got *tired* of it.

So we built





Logs, Metrics, Traces & Profiles. One backend. One binary. No trade-offs.

LICENSE

AGPLv3

Open source. Your data, your infrastructure.

STORAGE ENGINE

ClickHouse

Columnar. Compressed. S3 cold tier. Petabyte scale.

DEPLOY

Single binary

K8s, VM, bare metal. Fits your design. Production tested.

What Gigapipe **is** — and what it **isn't**.



✓ IS

- ✓ Drop-in compatible
- ✓ Protocol-native of industry standards
- ✓ Open source (AGPLv3)
- ✓ Single binary, Go
- ✓ ClickHouse-backed, Data Lake native in the next version
- ✓ Natively compatible with Grafana and Perses

✗ IS NOT

- ✗ A fork of the Grafana stack
- ✗ A proxy or middleware
- ✗ Vendor lock-in
- ✗ A rewrite of your agents
- ✗ Just another managed SaaS
- ✗ Opinionated about your setup

Gigapipe speaks *the* languages natively

INGESTION · WRITE

Push from anything.

- OpenTelemetry (OTLP)
- Loki Push API
- Prometheus Remote Write
- Tempo / Zipkin / Jaeger
- Pyroscope
- InfluxDB Line Protocol
- Datadog Agent API
- Elastic / OpenSearch

One storage backend.
One unified time index
across all signals.

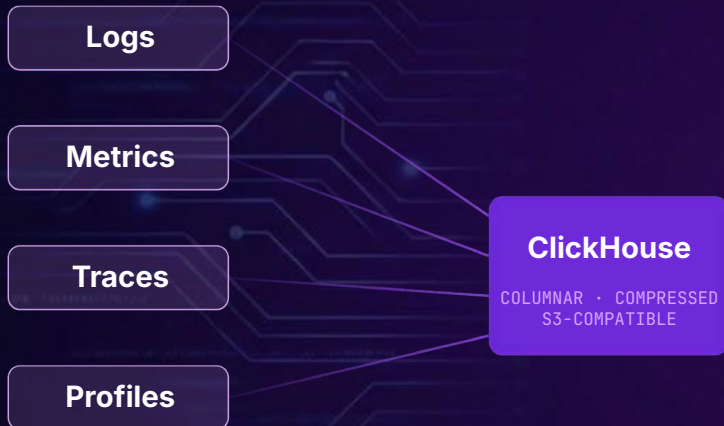
QUERY · READ

Read like anything.

- LogQL – Loki API
 - PromQL – Prometheus API
 - TraceQL – Tempo API
 - FlameQL – Pyroscope API
 - InfluxDB Query
 - ClickHouse SQL (native)
- NATIVE GRAFANA DATASOURCES

100% native implementation. Your agents and dashboards don't change. You change one endpoint.

Why ClickHouse isn't accidental



Compression on real telemetry



THE SHAPE FITS

Time-series with high-cardinality labels. Write-heavy, range-read, append-only. ClickHouse was built **exactly** for this.

COLUMNAR READS

"Logs from service X, last hour" touches only the columns it needs, not every row. That's why queries on months of raw data return in under a second.

S3 COLD TIER

Old telemetry moves to S3 automatically. Keep 90 days, 1 year. You're paying cents per GB, not a growing database bill.

SCALES WITH YOUR DATA

Start on a single node. Add nodes as data grows. No schema migration, no re-architecture, no downtime. Scale is a config change, not a project.

Modular architecture. That's it.



Ingestion Layer

OTel collector & protocols

OTel Collector

Custom Gigapipe

Drop-In Compatible

Protocol support built-in

NO Vendor Lock-In

Compatible w/ Datadog, Elastic, Influx

High Throughput

Billions events/day



Storage Layer

ClickHouse & S3

ClickHouse

columnar OLAP DB

S3 Compatible

Object storage for cold tier

Compression

High-fidelity retention

Data Sovereignty

User has full ownership & control



Query Layer

All 4 signal types

Clean Implementation

No Grafana code anywhere

All 4 signals

Logs, metrics, traces, profiles

Zero Retraining

Your team knows these already

Cross-Signal Correlation

Full context of your services



API & UI Layer

Endpoints & frontends

Native Query APIs

LogQL, PromQL, TraceQL, FlameQL

Grafana

OSS or Cloud

Perses

OSS CNCF project

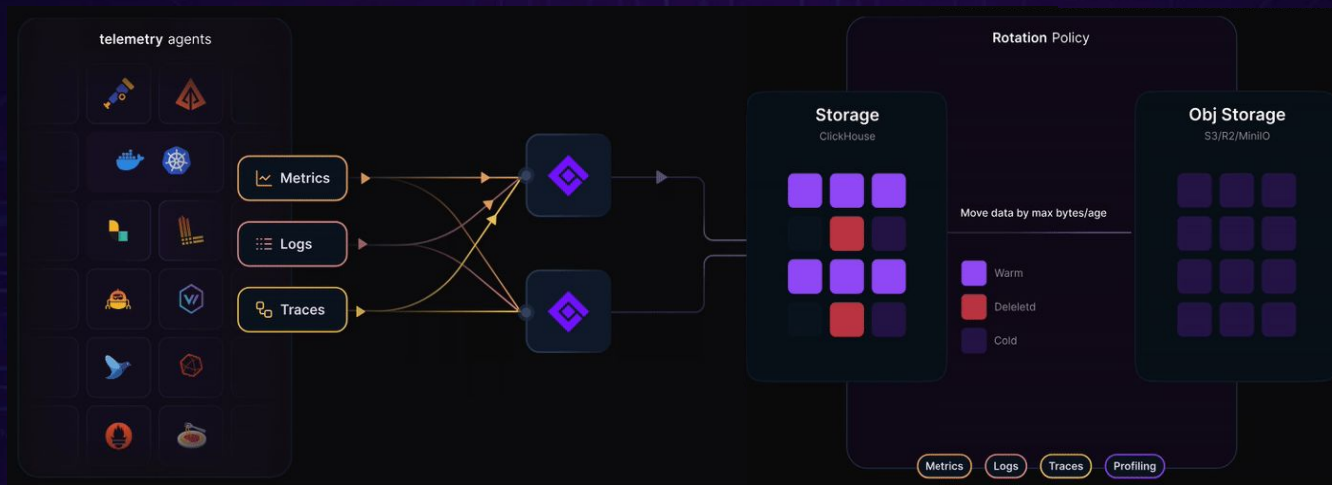
Gigapipe or any other

Any compatible UI.



Each layer is independent. Swap the UI without touching storage. Add a storage backend without changing agents. Nothing is coupled that doesn't need to be.

Why does it work



Columnar reads touch only what queries need. Fingerprint indexing reduces search space before data is scanned. Async inserts eliminate buffer middleware. The result: **sub-second queries on months of raw telemetry with no pre-aggregation.**



Ingestion Speed

Async inserts buffer server-side in ClickHouse, eliminating Kafka. Direct native protocol support means zero transformation overhead.



Storage Efficiency

ClickHouse MergeTree with LZ4 compression stores observability data at 10-20x better density than row-based databases.



Query Performance

Columnar reads touch only the columns needed. Vectorised execution processes data in CPU-cache-friendly batches.



Designed for Agents

Standard APIs mean any LLM, MCP, or agentic framework queries Gigapipe natively. No custom SDK. LogQL, PromQL, TraceQL, FlameQL — the same endpoints humans use, agents use too.

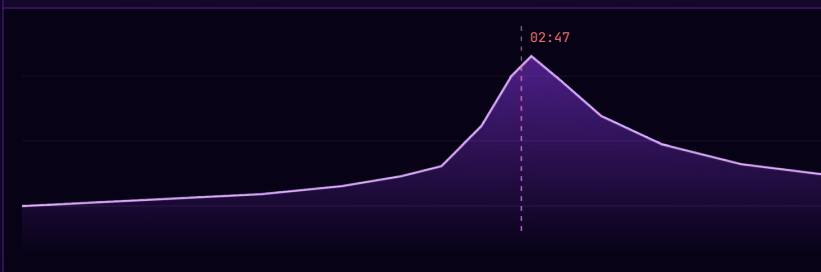
One tool. All four signals. Same time range.

GRAFANA · PAYMENT-SERVICE INCIDENT · last 15 min · datasource: Gigapipe

02:40 – 02:55 UTC

METRICS · http_req_p99

via Prometheus DS → Gigapipe



LOGS · {service="payment"} |> "ERROR"

via Loki DS → Gigapipe

```
02:47:13.221 svc=payment
ERROR payment timeout - trace=abc123
02:47:13.504 svc=payment
ERROR payment 5xx - trace=abc124
02:47:14.882 svc=payment
ERROR payment 5xx - trace=abc125
```

↑ click traceID → opens Tempo panel

TRACE abc123 · full span tree

via Tempo DS → Gigapipe



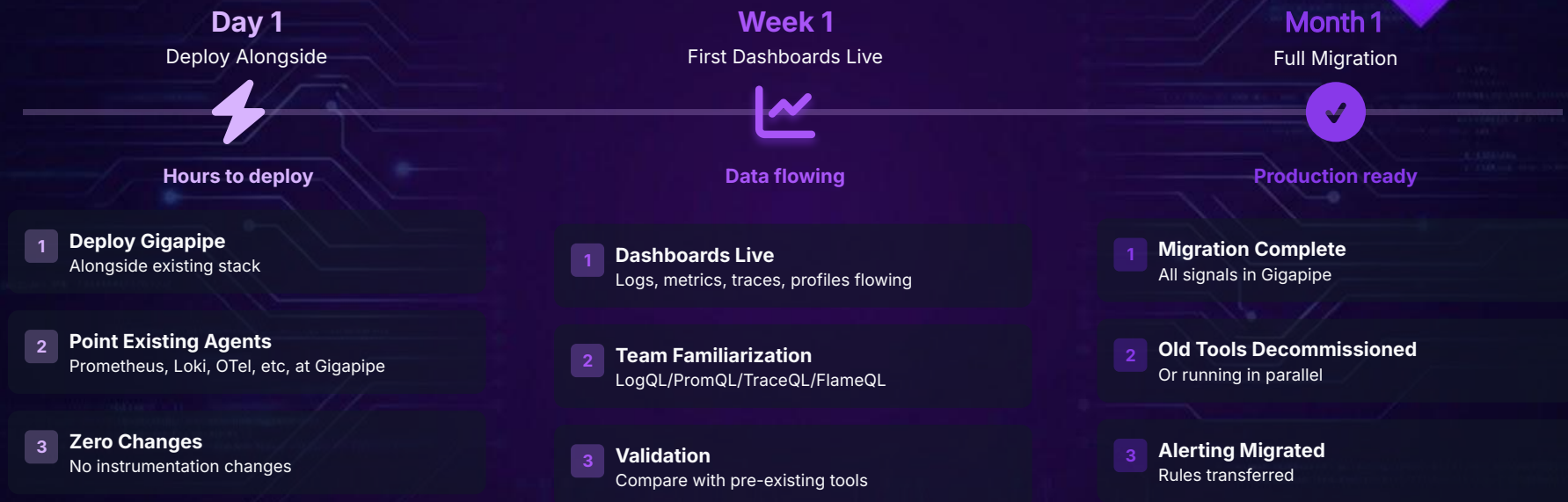
CPU PROFILE · payment-svc @ 02:47

via Pyroscope DS → Gigapipe



+15 min MTTR tax → 90 seconds. Manual correlation becomes a click.

From zero to production: One signal at a time



No professional services required for standard deployments. **No agent replacement.** Self-service deployment with documentation and community support.

In production. **Today.**



✓ **Single Go binary**

Kubernetes, VMs, bare metal — runs anywhere a Go binary runs.

✓ **Self-host or Cloud**

ClickHouse Cloud, your own cluster, or Gigapipe Cloud. Your data, your choice.

✓ **S3 cold tier**

Old telemetry moves to S3 automatically. You pay storage prices, not database prices.

✓ **Horizontal scale**

Stateless. Add nodes, add capacity. No migration. No downtime.

✓ **AGPLv3 open source**

Your data on your infrastructure. No license keys.

✓ **Active development**

Weekly releases. Maintainers respond. GitHub and Matrix.

PRODUCTION NUMBERS

Billions

data points ingested daily

across production deployments today

<1s

query response on live data

no pre-aggregation, no caching tricks

50–90%

cost reduction

vs Datadog, New Relic, Elastic at equivalent scale

● INCIDENT · RESOLVED

It's 3am.

Your phone is ringing.

One tool. Four signals. +15 min → 90 seconds.

Run it tonight.



~/your-stack · SRE DAY

```
$ docker run -p 3100:3100 -p 8123:8123 \
ghcr.io/metrico/gigapipe:latest
```

```
# point your existing Loki / Prom / OTel agent at it
# your Grafana datasources just work – verify in one dashboard
```



Star it on GitHub

github.com/metrico/gigapipe



Tell your team

show them these slides



Contribute

[issues](#) · [docs](#) · [new protocols](#)

COMMUNITY

Matrix
#qryn:matrix.org

Maintainers respond.

DOCS

gigapipe.com/docs

Architecture, deployment, all protocols.

TRY CLOUD

gigapipe.com

Free trial. Managed by us.

*The stack we wished for... **Now exists!***



Thank you.

Questions?



info@gigapipe.com



gigapipe.com



github.com/metrico/gigapipe